E-Commerce Pricing Strategy A/B Test Analysis

Comparing Two Pricing Strategies Using Statistical Hypothesis Testing

MINI PROJECT REPORT

Project Focus Evaluation of a new pricing / landing page against the existing page.
Dataset 294,478 original A/B-test records; 290,584 unique users after cleaning.
Primary Finding No statistically significant improvement in conversion from the treatment page.
Analyst Tanushka Bhadouriya

Executive Summary

This project evaluates whether a new e-commerce pricing/landing page, referred to as the treatment, converts visitors better than the existing control page. After removing invalid group/page assignments and duplicate users, 290,584 unique users were analyzed across Canada, the United Kingdom, and the United States.

The control conversion rate was 12.039%, compared with 11.881% for treatment. The observed difference was -0.158 percentage points. The primary two-proportion Z-test

returned Z = -1.3109 and p = 0.1899, which is above the 0.05 significance threshold. A country-adjusted logistic regression produced a treatment p-value of 0.1912 and an odds ratio of 0.9852.

Overall conclusion: the experiment does not provide statistically significant evidence that the new pricing page improves conversion. The existing page should therefore be retained unless there is another business reason to change it.

Contents

1. Introduction

2. Data Description

2.1 Data Cleaning

3. Methodology

3.1 Two-Proportion Z-Test

3.2 Country-Adjusted Logistic Regression

3.3 Country Segmentation

3.4 Bayesian A/B Analysis

4. Results

4.1 Descriptive Overview

4.2 Primary Z-Test Results

4.3 Logistic Regression Results

4.4 Country-Level Results

5. Discussion

6. Conclusion and Recommendation

1 Introduction

E-commerce companies routinely test changes to pricing and landing pages before deploying them to every customer. A/B testing allows the business to compare two versions under controlled conditions and determine whether the observed difference is large enough to be considered statistically meaningful.

In this experiment, visitors were assigned to two groups. The control group was shown the existing old page, while the treatment group was shown a new page. For each observation, the dataset records whether the user converted, where converted = 1 represents a conversion and converted = 0 represents no conversion.

Business question: Does the new pricing strategy produce a higher conversion rate than the existing strategy with statistical confidence, or could the observed difference reasonably be explained by random variation?

The analysis uses conversion rate as the primary performance metric. The main statistical test is a two-proportion Z-test because the outcome is binary. A country-adjusted logistic regression and country-level segmentation are used as additional checks.

2 Data Description

Two files were used for the analysis. The first contains the experimental observations and the second maps users to their country.

Dataset	Rows	Variables / Purpose
A/B test data	294,478	user_id, timestamp, group, landing_page, converted
Country data	290,584	user_id, country

2.1 Data Cleaning

The raw A/B-test data contained two important quality issues. First, some observations had a mismatch between the experimental group and the page displayed. A control user should have been assigned to old_page, while a treatment user should have been assigned to new_page. The analysis identified 3,893 mismatched records and removed them.

Second, duplicate user IDs were present. Because the experiment should count each user once, the data was sorted by timestamp and the first observation for each user was retained. The analysis identified 3,894 duplicate user records.

After cleaning and merging the country table, the final dataset contained 290,584 unique users. The groups were almost perfectly balanced: 145,274 control users and 145,310 treatment users.

3 Methodology

All hypothesis tests use a significance level of $\alpha = 0.05$. The null hypothesis represents no treatment effect. A p-value below 0.05 is interpreted as statistically significant evidence against the null hypothesis.

3.1 Two-Proportion Z-Test

$\mathbf{H_0}$: $p_{treatment} = p_{control}$
$\mathbf{H_1}$: $p_{treatment} \neq p_{control}$

The two-proportion Z-test is the primary test because conversion is a binary outcome. It directly compares the conversion proportions of the treatment and control groups. A confidence interval is also calculated to quantify the plausible range of the difference.

3.2 Country-Adjusted Logistic Regression

A logistic regression model was used to determine whether the treatment effect remains after accounting for country. The model contains treatment assignment and country as predictors of conversion. The treatment coefficient is transformed into an odds ratio, which indicates the multiplicative change in the odds of conversion.

3.3 Country Segmentation

Conversion rates were calculated separately for Canada, the United Kingdom, and the United States. This checks whether the overall result hides a country-specific pattern where the new page performs substantially better or worse.

3.4 Bayesian A/B Analysis

A Bayesian comparison was also used to estimate the probability that the treatment conversion rate is higher than the control conversion rate. The supplied analysis reports a probability of 9.59%, providing an additional interpretation of the evidence.

4 Results

4.1 Descriptive Overview

The final dataset contains 290,584 unique users. The control group has 145,274 users and 17,489 conversions, while the treatment group has 145,310 users and 17,264 conversions.

Group	Users	Conversions	Conversion Rate
Control	145,274	17,489	12.039%
Treatment	145,310	17,264	11.881%
Difference	—	—	-0.158 pp

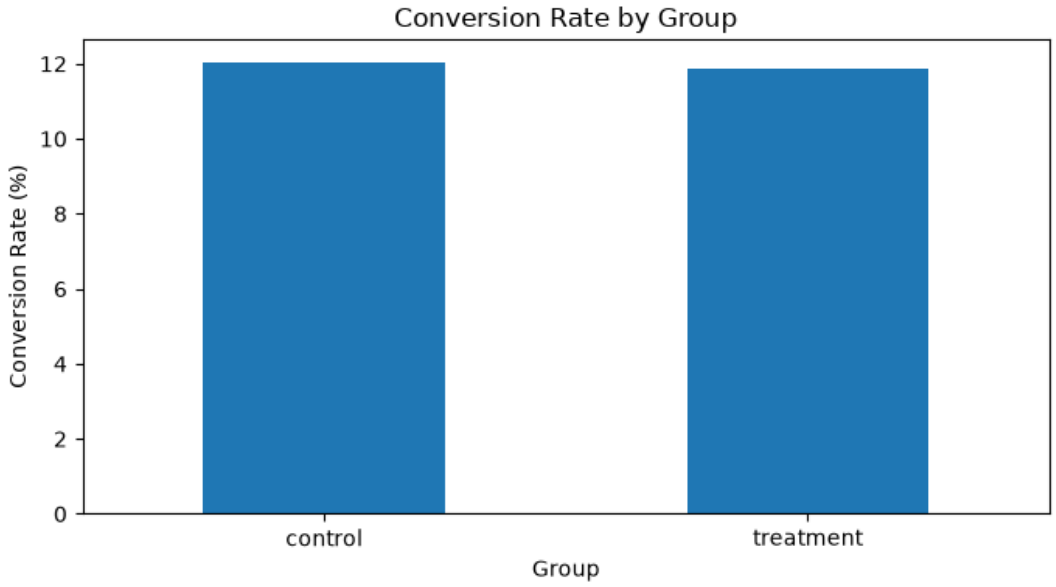


Figure 1. Conversion rate by experimental group. The control group is slightly higher, but the observed difference is small.

4.2 Primary Z-Test Results

Metric	Result
Control conversion rate	12.039%
Treatment conversion rate	11.881%
Observed difference	-0.158 percentage points
Z-statistic	-1.3109
p-value	0.1899

Metric	Result
Control 95% CI	11.871% – 12.206%
Treatment 95% CI	11.714% – 12.047%
Decision	Fail to reject H0

Because p = 0.1899 is greater than 0.05, the null hypothesis is not rejected. There is no statistically significant evidence that the treatment page changes conversion rate. The treatment is numerically lower by 0.158 percentage points, but this difference is compatible with sampling variation.

4.3 Logistic Regression Results

Term	Coefficient	p-value
Intercept	-2.0300	<0.001
Country: UK	0.0506	0.0745
Country: US	0.0408	0.1295
Treatment	-0.0149	0.1912

The treatment coefficient is -0.014943 and the corresponding odds ratio is 0.9852. This indicates that the estimated treatment odds of conversion are approximately 1.48% lower than control after controlling for country. However, p = 0.1912, so the adjusted treatment effect is not statistically significant.

4.4 Country-Level Results

Country	Group	Users	Conversion Rate
CA	Control	7,198	11.878%
CA	Treatment	7,301	11.190%
UK	Control	36,360	12.002%
UK	Treatment	36,106	12.117%
US	Control	101,716	12.063%
US	Treatment	101,903	11.847%

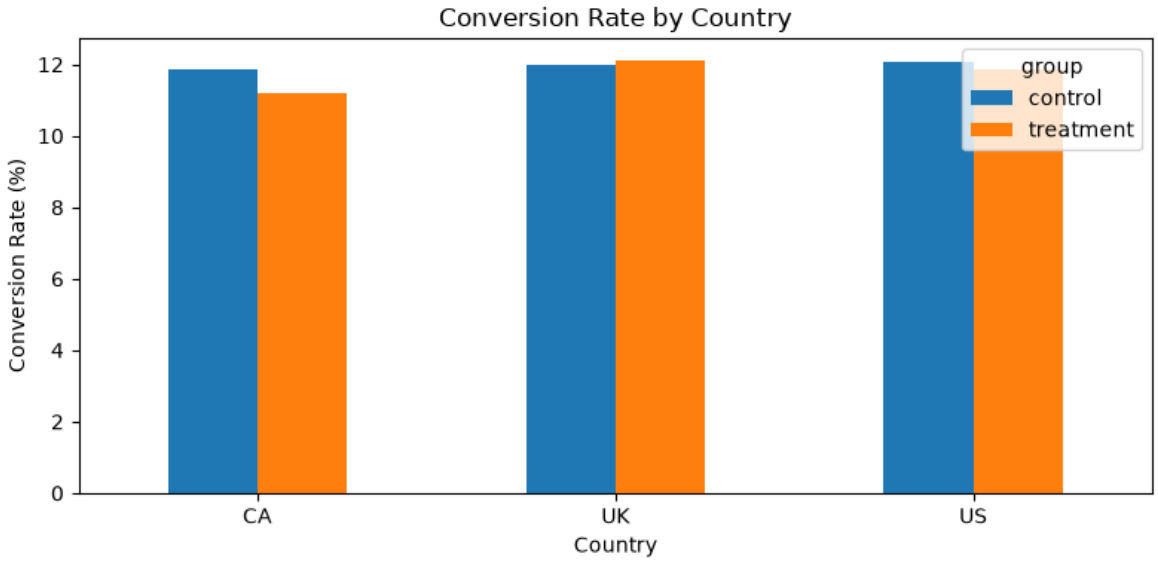


Figure 2. Conversion rate by country and experimental group. The treatment does not show a consistent advantage across countries.

In Canada, treatment conversion is lower than control. In the UK, treatment is slightly higher. In the US, treatment is again lower. The direction therefore changes across countries rather than showing a consistent treatment advantage.

5 Discussion

The primary and adjusted analyses produce the same overall conclusion. Control conversion is 12.039%, while treatment conversion is 11.881%. The observed difference of -0.158 percentage points is small, and the primary Z-test gives p = 0.1899. The country-adjusted logistic regression gives p = 0.1912.

The Bayesian analysis also does not support the treatment: the estimated probability that treatment performs better than control is only 9.59%. The expected absolute lift is

approximately -0.158 percentage points.

The country-level results do not identify a hidden segment where the treatment clearly wins. CA and US favor control, while UK shows only a very small numerical advantage for treatment.

A non-significant result should not be interpreted as proof that the two pages are identical. It means that the available experiment does not provide sufficient statistical evidence of a difference. Very small true effects can still be difficult to distinguish from random variation.

6 Conclusion and Recommendation

Based on the analyzed data, there is **no statistically significant evidence that the new pricing page improves conversion**. The treatment page actually has a slightly lower observed conversion rate, but the difference is not statistically significant.

RECOMMENDATION
Retain the current pricing page rather than switching to the new page solely on the basis of this experiment. The treatment has not demonstrated a measurable conversion benefit.
If the business expects a very small improvement, conduct a follow-up experiment designed with sufficient statistical power to detect that smaller effect. A more substantially redesigned pricing experience may also be worth testing.

Key results: Control = 12.039% | Treatment = 11.881% | Difference = -0.158 pp | Z = -1.3109 | p = 0.1899 | Adjusted p = 0.1912 | Odds ratio = 0.9852 | P(Treatment > Control) = 9.59%.

Source: user-provided A/B test analysis outputs and the supplied sample report.